SPHEREx Motion Explorer

A public workbench that turns the official IRSA archive into an interactive time-domain sky experience: registered epoch comparisons, on-the-fly spectra, and Gaia proper-motion overlays - all verified live against real mission data.

SPHEREx Motion Explorer team - v0.5.0 -

November 2026

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Project pitch - NASA Space Apps Challenge 2026 - v0.5.0, November 2026

1. The Challenge

Since May 2025, NASA's SPHEREx mission has mapped the entire sky every six months in 102 bands of near-infrared light, returning images of more than a billion objects. Comets, asteroids, brown dwarfs and planets reveal themselves by **shifting position between images** - the same technique that discovered Pluto in 1930. No single person can sift through that volume of data. The Space Apps challenge asked for a public-facing web tool that displays SPHEREx images of any patch of sky and makes temporal changes easy for anyone to see.

102	1B+	6 mo	0
infrared bands per sky map	objects in the SPHEREx archive	all-sky mapping cadence	images simulated or faked

2. Our Answer

SPHEREx Motion Explorer is an observational workbench that turns the official NASA/IPAC IRSA archive into an interactive time-domain experience. Enter any sky coordinate, retrieve every compatible real SPHEREx exposure, and compare epochs as registered, blink-comparable frames. The tool adds the scientific glue the raw archive lacks: per-pixel wavelength awareness, bit-correct quality masking, on-the-fly spectrophotometry, and a Gaia proper-motion overlay that flags fast movers automatically.

130	37 d	40/40	10.4"/yr
real QR3 records retrieved for Barnard's Star	verified temporal baseline	spectrum points extracted, 0 failures	proper motion auto-detected by the overlay

3. The Judging Journey - Six Steps

Step	Action	What actually happens (verified live)
01	Choose a field	Five preset challenge targets (Barnard's Star, epsilon Indi B, WISE 0855-0714, RW Aurigae, R Draconis) or any RA/Dec. Field of view 1-18 arcmin, QR2/QR3, wide or deep collections, UTC filters.

Step	Action	What actually happens (verified live)
02	Retrieve real observations	The app queries IRSA SIA v2 live. 130 dated records returned for Barnard’s Star; every record shows its detector, wavelength range, and off-axis angle so users pick frames that truly contain the target.
03	Compare epochs	One click selects the widest target-centred baseline. Blink, split, mean, and exploratory B-minus-A modes share one display stretch so brightness changes are not masked by per-frame contrast.
04	See the physics	Each frame reports its LOCAL wavelength at the target from the spectral WCS - the tool exposed a 0.50 micron mismatch inside one detector band, and a 0.23 micron difference between two epochs of the same band.
05	Extract a spectrum	The on-the-fly extractor sweeps up to 40 archive files in parallel and builds a 0.8-4.8 micron spectrum with per-point statistical errors, detector colour-coding, and CSV export.
06	Overlay the sky	Gaia DR3 stars (epoch-propagated) ring every field star; any source moving faster than 1 arcsec/yr gets an amber ring, a motion arrow, and a measured speed label. A two-layer AllWISE atlas puts the field in mid-infrared context.

4. Built for Trust: Science Integrity by Design

Most astronomy demo tools break quietly on real data. Ours publishes its own audit trail. The flag-masking system follows the official SPHEREx Explanatory Supplement bit table - critically, bit 21 (SOURCE) marks pixels containing real star flux and is never masked, while genuinely fatal statuses (saturation, non-linearity, persistence, satellite streaks) are. A naive mask deleted 96 percent of pixels and rendered Barnard’s Star invisible; ours restores 95 percent valid coverage. Every rendered frame carries a provenance header, every spectrum point is labelled statistical-only, and every screen states what the tool is not: it does not certify discoveries.

Deliberate honesty feature	Why it matters
Honest empty states	No coverage means an explicit message - invented sky data can never appear
Local wavelength matching	Same detector does not mean same band; the tool quantifies the mismatch
Saturation-aware spectra	Bright-source points are flagged as lower limits, never silently wrong
Evidence manifests	One JSON export preserves observation IDs, caveats, DOI, and a reproducible URL
Residual cues, not detections	Difference-mode highlights are review aids, explicitly not claims

5. Architecture at a Glance

A dependency-light stack chosen so any volunteer can run it: vanilla HTML/CSS/JavaScript frontend (no build step), a FastAPI backend that brokers every archive call, Astropy + reproject for FITS science, Aladin Lite v3 vendored locally (no CDN dependency), and Docker packaging. Everything talks to the official NASA/IPAC IRSA services - SIA v2 for discovery, IBE for cutouts and AllWISE reference imagery, TAP for Gaia.

Layer	Technology	Role
Frontend	Vanilla JS + Aladin Lite v3 (vendored)	Workbench UI, WCS canvas overlays, zoom/pan, spectrum plot
Backend	Python FastAPI + httpx	Archive brokering, retries, SSRF allowlist, bounded caches
Science	Astropy WCS + reproject + numpy	Flag masking, reprojection, aperture photometry, mini-FITS generation
Data	NASA/IPAC IRSA (SIA, IBE, TAP)	Exposure discovery, FITS cutouts, AllWISE atlas, Gaia DR3
Delivery	Docker + Cloudflare tunnel / any host	Public URL without infrastructure cost

6. Verified, Not Claimed

Every number in this document was measured against the live archive on the submission day. The pipeline runs end-to-end on all five challenge targets; Barnard's Star and epsilon Indi B were exercised most deeply.

95%
valid pixels restored by bit-correct masking

0.8-4.8 um
verified spectrum coverage, all 6 detectors

SNR 175-406
per-point signal-to-noise on Barnard's Star

300
Gaia reference stars overlaid per field

7. Who It Serves

- **Citizen scientists** - a zero-install browser workbench that mirrors professional blink-comparison workflows, with honest guidance at every step.
- **Educators and students** - real mission data with the physics (variable filter, saturation, proper motion) made visible instead of hidden.
- **Researchers** - a fast triage layer over IRSA: wavelength-matched epoch selection, quick spectra, and evidence exports reproducible from archive IDs.

8. Roadmap

Horizon	Item
Next	Season-aware SIA pagination beyond 600 records; PSF-matched difference photometry using the EPSF extension
Next	Official IRSA spectrophotometry cross-checks and uncertainty plots beside our exploratory spectrum
Later	Solar-system object path overlays (MOST) so known asteroids are never mistaken for discoveries

Horizon	Item
Later	Multi-tile AllWISE mosaics beyond the 1.25-degree context limit; hosted always-on deployment

9. Try It

- **Public URL:** live demo tunnel (printed by start_public.bat on the project machine)
- **Source code:** github.com/shree-aru/spherex-motion-explorer
- **Local run:** double-click run_windows.bat, open http://127.0.0.1:8000
- **Data credit:** SPHEREx data are served live from NASA/IPAC IRSA; the tool is an independent citizen-science application, not official NASA software.

Built with enthusiasm by a team that believes the sky belongs to everyone - and that honest tools make better discoveries.